

Initial Project Planning Template

Date	15 March 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	User Registration	USN-1	As a farmer, I can register in the application using email and password.	2	High	Team	28-06-2026	30-06-2026
Sprint-1	Login	USN-2	As a user, I can log in securely to access the system.	2	High	Team	28-06-2026	30-06-2026
Sprint-2	Soil Data Input	USN-3	As a farmer, I can enter soil data like Nitrogen, Phosphorus, Potassium and pH values.	3	High	Team	01-07-2026	04-07-2026
Sprint-2	Weather Data Input	USN-4	As a farmer, I can enter temperature, humidity and rainfall details.	2	High	Team	01-07-2026	04-07-2026
Sprint-3	Crop Recommendation	USN-5	System predicts the most suitable crop based on input data.	5	High	Team	05-07-2026	09-07-2026
Sprint-3	ML Model Training	USN-6	Train machine learning model for accurate crop prediction.	5	High	Team	05-07-2026	09-07-2026
Sprint-4	Dashboard	USN-7	Display crop recommendation and results in dashboard.	3	Medium	Team	10-07-2026	13-07-2026
Sprint-4	Notification System	USN-8	Send recommendations and alerts to users.	2	Medium	Team	10-07-2026	13-07-2026
Sprint-5	Testing	USN-9	Test system accuracy, performance and reliability.	4	High	Team	14-07-2026	17-07-2026
Sprint-5	Deployment	USN-10	Deploy the project for final demonstration.	3	Medium	Team	14-07-2026	17-07-2026

Initial Project Planning Phase

The Initial Project Planning phase focuses on organizing the development process of the **Smart Agricultural Production Optimization Engine (OptiCrop)**. This phase helps define project milestones, functional requirements, sprint schedules, and task estimation to ensure smooth project execution.

The project follows an iterative development approach where the complete system is divided into multiple sprints. Each sprint focuses on delivering specific functionalities such as user registration, soil data collection, weather data input, crop recommendation, machine learning model training, dashboard development, testing, and deployment.

By maintaining a structured product backlog and sprint schedule, the team can efficiently track progress, manage resources, and complete the project within the planned timeline.

Project Objectives

- Develop a smart crop recommendation system
- Collect soil and environmental data from farmers
- Analyze data using machine learning algorithms
- Recommend suitable crops for maximum yield
- Improve farming productivity and resource optimization

Sprint Planning Objectives

Sprint 1

- User Registration
- Login Authentication

Sprint 2

- Soil Data Input
- Weather Data Collection

Sprint 3

- Crop Prediction
- ML Model Training

Sprint 4

- Dashboard Development
- Notification System

Sprint 5

- Testing and Deployment

Expected Deliverables

- Functional web application
- Machine learning crop prediction model
- Interactive dashboard
- Testing report
- Final deployment for demonstration

Benefits of Planning

- Better task management
- Efficient resource allocation
- Faster development
- Clear project roadmap
- Improved team collaboration